

Exercise series 5: SQL and Aggregation

Introduction to databases



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The employees database schema is defined as:

- **employees**(emp_no, birth_date, first_name, last_name, gender, hire_date)
- **departments**(dept_no, dept_name)
- **dept_manager**(emp_no, dept_no, from_date, to_date)
- **dept_emp**(emp_no, dept_no, from_date, to_date)
- **titles**(emp_no, title, from_date, to_date)
- **salaries** (emp_no, salary, from_date, to_date)

- Note that most tables contain many (thousands or millions) of records, so using **LIMIT** is important.
- If you need to use the current date you can use the function **NOW()** that outputs the current date and time.
- You can use the **DROP** command to remove a table.
- You can use the **OR REPLACE VIEW** command to replace an existing view

```
SELECT * FROM departments LIMIT 10;
```

```
SELECT * FROM dept_emp WHERE to_date > NOW();
```

```
DROP TABLE test_table;
```

```
CREATE OR REPLACE VIEW test_view AS [...];
```

Write the following queries in SQL

1. What is the name of the last employee in the database?
2. List the department he works for?
3. What is the name of his current boss?
4. What is his current salary?
5. List the number of employees working in each department.
6. List all departments and the name of the current manager.
7. Find the names of managers that earn less than the highest paid manager.
8. Find all employees that have worked for at least two departments.
9. Find all employees that had the same title during their career.
10. ! Find all employees that had all possible titles during their career.
11. ! Find all employees that had an increase, between their first and current salary, that is at least 2 times the average increase in salary for any employee in the same department.
12. Find the top-10 employees with the highest last salary
13. Find for each department the total, maximal, average and minimal salary.

